

Question	Answer	Marks
6(a)	capacitance = charge / potential	M1
	charge is (numerically equal to) charge on one plate	A1
	potential is potential difference between plates	A1
6(b)(i)	two in series, in parallel with the other (correct symbols)	A1
6(b)(ii)	two in parallel connected to one in series (correct symbols)	A1
6(c)(i)	capacitance = $1.2 \mu\text{F}$	A1
6(c)(ii)	1. $Q = CV$	C1
	$= 1.2 \times 8.0$ $= 9.6 \mu\text{C}$	A1
	2. $E = \frac{1}{2}QV$ and $V = Q/C$ or $E = \frac{1}{2}CV^2$ and $V = Q/C$ or $E = \frac{1}{2}Q^2/C$	C1
	$E = \frac{1}{2}(9.6 \times 10^{-6})^2 / (3.0 \times 10^{-6})$ $= 1.5 \times 10^{-5} \text{ J}$	A1

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7(c)(i)	(fraction of) output is combined with the input	M1